# Super-Convergence: Very Fast Training of Neural Networks Using Large Learning Rates

Разбор подготовил Заволович Богдан

# Intro

This paper studies how the learning rate policy affects both training speed and generalization.

$$\theta_{i+1} = \theta_i - \epsilon \bigtriangledown_\theta f(\theta_i)$$

The learning rate scales the parameter update at iteration t

# The cost of standard training

Baseline: a piecewise constant learning rate, reduced at accuracy plateau.

ResNet-56 on CIFAR-10 takes about 80 000 iterations to reach 91.2% accuracy

Larger learning rates can speed up training, but may cause instability of training and divergence.

# Question

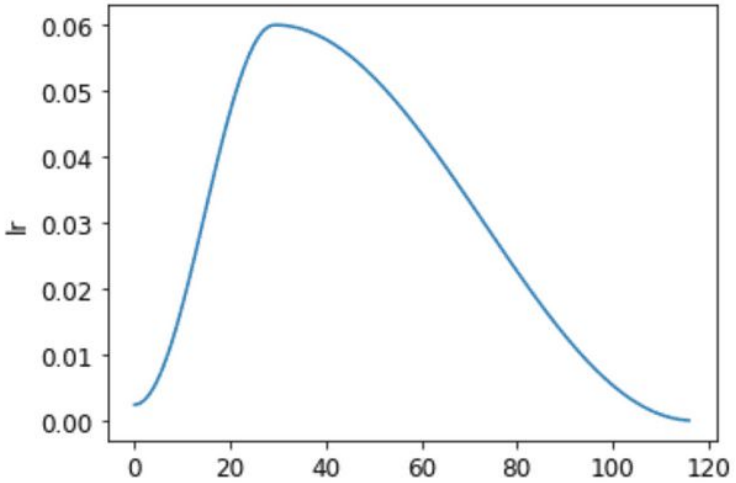
Can we train a network in far fewer steps while preserving or even improving it’s generalization capabilities?

# The 1cycle learning rate policy

Increase the learning rate

Reduce the learning rarte after the peak

Finish well below the initial learning rate



# Large LR as regularization

As the learning rate rises, training loss can increase while test loss decreases

The authors interpres this as a regularization effect

# Choosing the learning rate range

Increase the learning rate gradually and track loss and accuracy

High accuracy across a wide learning rate range shows potential for super convergence.

Tune the learning rate together with weight decay, batch size, momentum and batch norm.

# Resnet56 10 000 vs 80 000 iterations

8 times less iterations, with a 1.2% accuracy gain

| Training setup | Iterations | Test accuracy |
|---|---|---|
| Piecewise constant LR | 80000 | 91.2% |
| CLR, LR range 0.1 - 3 | 10000 | 92.4% |

# Results across architectures and datasets

| Model / Dataset | Epochs<br>Baseline / 1cycle | Baseline accuracy (%) | 1cycle accuracy (%) |
|---|---|---|---|
| Wide Resnet / CIFAR10 | 800/100 | 90.3 | 91.9 |
| Densenet / CIFAR10 | 400/150 | 92.7 | 92.8 |
| LeNet / MNIST | 85/12 | 99.03 | 99.25 |

# Limitations and takeaway

Result – fewer iterations with comparable or higher test accuracy

Learning rate policy and regularization need to be tuned together